Contingency Table (2)
(분할표 (2))

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Today's Topics

1. Test of independence
   - Pearson Chi-square statistic
   - Measure of Association
   - Test for Independence for Ordinal Data
   - Fisher's Exact Test
2. Partial Association
   - Three-way Table
   - Conditional Association
   - Homogeneous Association

Chi-square Test for Independence

Chi-square Distribution

Distribution Theory:

If $Z \sim N(0,1)$,then $Z^2 \sim \chi^2_1$

If $X_1 \sim \chi^2_{v_1} \perp X_2 \sim \chi^2_{v_2}$, then $X_1^2 + X_2^2 \sim \chi^2_{v_{1+}v_2}$

If $t \sim t_v$, then $t^2 \sim F_{1,v}$

$E\left(\chi^2_{df}\right) = df$ and $Var\left(\chi^2_{df}\right) = 2df$

$\chi^2_{df} \overset{\mathrm{d}}{\equiv} \Gamma(\frac{df}{2}, 2)$

Likelihood Ratio Test (LRT)

Parameter space (Ω): the complete set of values that a parameter can take

(Example) For the binomial distribution $X \sim B(n, \pi)$

$$\Omega = \{(n, \pi) | n = 1, 2, \ldots, 0 \leq \pi \leq 1\}$$

(Example) For the Poisson distribution $X \sim P(\lambda)$

$$\Omega = \{\lambda | \lambda > 0\}$$

Partitioning of the parameter space

Ω_0: parameter space under the null hypothesis

Ω_1: parameter space under the alternative hypothesis

Note: $\Omega = \Omega_0 \cup \Omega_1$ and $\Omega_0 \cap \Omega_1 = \emptyset$

Likelihood Ratio Test (LRT) (Cont'd)

Likelihood Function:

For a given observed sample, the likelihood function is expressed as:

$$L(\theta) = f(x_1, x_2, \dots, x_n; \theta)$$

, which represents the probability (or density) of observing the data as a function of the paremeter θ

Maximum Likelihood Estimate (MLE):

The value of θ that maximizes $L(\theta)$

Likelihood Ratio Test (LRT) (Cont'd)

Likelihood Ratio Test:

$$H_0:\ \theta_1 = \theta_{10} \ \text{vs}\ H_1:\ \theta_1 \neq \theta_{10}\ ,\ \theta = (\theta_1, \theta_2)^t,\ \theta_1 \in R^{n_1}, \theta_2 \in R^{n_2}$$

$L(\hat{\theta}_0)$: MLE under H_0

$L(\hat{\theta}_A)$: MLE under Ω

Test statistic:

$$\Lambda = \frac{L(\widehat{\theta}_0)}{L(\widehat{\theta}_A)}, \ -2\log\Lambda \sim \chi^2(df = n_1)$$

Properties:

$0 < \Lambda \leq 1$

Large values of Λ support H_0, small values support H_1

Wald Test

$$H_0\colon\ \theta_1 = \theta_{10}\ \text{ vs }\ H_1\colon\ \theta_1 \neq \theta_{10}\ \ ,\ \ \theta = (\theta_1, \theta_2)^t,\ \ \theta_1 \in R^{n_1}, \theta_2 \in R^{n_2}$$

$\hat{\theta}_0 = \left(\theta_{10}, \hat{\theta}_{20}\right)$: MLE under H_0

$\hat{\theta} = \left(\hat{\theta}_1, \hat{\theta}_2\right)$: MLE under Ω

$$I_{1|2}(\hat{\theta}) = I_{11}(\hat{\theta}) - I_{12}(\hat{\theta})[I_{22}(\hat{\theta})]^{-1} I_{21}(\hat{\theta}),\ \ I(\theta) \equiv \begin{pmatrix} I_{11}(\theta) & I_{12}(\theta) \\ I_{21}(\theta) & I_{22}(\theta) \end{pmatrix},\ \ I_{ij}(\theta) = \frac{\partial^2 l}{\partial \theta_i \partial \theta_j}$$

Test statistics:

$$\left(\hat{\theta}_1 - \theta_{10}\right)^t I_{1|2}(\hat{\theta}) \left(\hat{\theta}_1 - \theta_{10}\right) \sim \chi^2(df = n_1)$$

Score Test

$$H_0:\ \theta_1 = \theta_{10}\ \text{vs}\ H_1:\ \theta_1 \neq \theta_{10}\ ,\ \theta = (\theta_1, \theta_2)^t,\ \theta_1 \in R^{n_1}, \theta_2 \in R^{n_2}$$

$\hat{\theta}_0 = \left(\theta_{10}, \hat{\theta}_{20}\right)$: MLE under H_0

$\hat{\theta} = \left(\hat{\theta}_1, \hat{\theta}_2\right)$: MLE under Ω

$$I_{1|2}(\hat{\theta}) = I_{11}(\hat{\theta}) - I_{12}(\hat{\theta})[I_{22}(\hat{\theta})]^{-1} I_{21}(\hat{\theta}),\ \ I(\theta) \equiv \begin{pmatrix} I_{11}(\theta) & I_{12}(\theta) \\ I_{21}(\theta) & I_{22}(\theta) \end{pmatrix},\ \ I_{ij}(\theta) = \frac{\partial^2 l}{\partial \theta_i \partial \theta_j}$$

$$S(\theta) \equiv (S_1(\theta), S_2(\theta)),\ S_1(\theta) \equiv \frac{\partial l}{\partial \theta_1},\ S_2(\theta) \equiv \frac{\partial l}{\partial \theta_2}$$

Test statistics:

$$S_1(\hat{\theta}_0)^t [I_{1|2}(\hat{\theta}_0)]^{-1} S_1(\hat{\theta}_0) \sim \chi^2(df = n_1)$$

Score Test (Cont'd)

(Example) Suppose $Y_1, \dots, Y_n \sim N(\mu, \sigma^2)$, derive the score statstics for:

$$H_0\colon \mu = \mu_0 \text{ vs } H_1\colon \mu \neq \mu_0$$

$$\frac{n(\bar{Y} - \mu_0)^2}{\frac{1}{n}\sum(y_i - \mu_0)^2}$$

Test of goodness

Suppose $X \sim B(n, \pi)$, consider hypothesis testing for $H_0: \pi = \pi_0$ (when n is large)

$$Z = \frac{p - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}} \sim N(0,1)$$

(i) $Z^2 \sim \chi^2(df = 1)$

(ii) $Z^2 = \frac{(X - n\pi_0)^2}{n\pi_0(1-\pi_0)} = \frac{(X - n\pi_0)^2}{n\pi_0} + \frac{(n - X - n(1-\pi_0))^2}{n(1-\pi_0)}$

$$= \sum_{i=1}^{2} \frac{(obesrved_i - Expected_i)^2}{Expected_i}$$ (Pearson chi-square test)

Test of goodness (Cont'd)

Suppose $(X_1, \dots, X_m) \sim M(n, \pi_1, \dots, \pi_{m-1})$ (multinomial distribution)

consider hypothesis testing for H_0: $\pi_1 = \pi_{01}, \dots, \pi_{m-1} = \pi_{0(m-1)}$ (when n is large)

$$X^2 = \sum_{i=1}^{m} \frac{(X_i - n\pi_{0i})^2}{n\pi_{0i}} \sim \chi^2(m-1)$$

Pearson Chi-square Statistic

Score test for $H_0 : \pi_{ij} = \pi_{i+}\pi_{+j}, \ \forall\, i,j \ \text{vs} \ H_1 : \text{not} \ H_0$

$$X^2 = \sum \frac{(n_{ij} - \mu_{ij})^2}{\mu_{ij}} \sim \chi^2(I-1)(J-1)$$

The larger the difference between $\{n_{ij}\}$ and $\{\mu_{ij}\}$,the stronger the evidence to reject H_0

The p-value is the probability that the chi-square statistic is greater than or equal to the observed value

Greater weight is assigned to cells with smaller μ_{ij}

Requires the condition $\{n_{ij} \geq 5\}$

For large samples, the statistic is approximately chi-square distributed

At least 75% of the cells should have expected values greater than 5

Pearson Chi-square Statistic (Cont'd)

Likelihood ratio test for $H_0 : \pi_{ij} = \pi_{i+}\pi_{+j},\ \forall\, i,j$ vs $H_1 :$ not H_0

$$G^2 = 2\sum_{i,j} n_{ij} \log\left(\frac{n_{ij}}{\mu_{ij}}\right) \sim \chi^2(df = (I-1)(J-1))$$

$$G^2 = -2\log(\Lambda)$$

$0 \le G^2 < \infty$,and larger values provide stronger evidence against the null hypothesis

If $n_{ij} = \mu_{ij}$ for all cells, then $G^2 = 0$

The relationship between χ^2 and G^2

$$o\log\left(\frac{o}{e}\right) = (o-e) + \frac{(o-e)^2}{2e} + \cdots$$

$$G^2 = 2\sum_{ij} (n_{ij} - \mu_{ij}) + \frac{(n_{ij}-\mu_{ij})^2}{2\mu_{ij}} + \cdots \simeq \sum_{ij} \frac{(n_{ij}-\mu_{ij})^2}{\mu_{ij}}$$

Test of independence

Statistically independent: $H_0 : \pi_{ij} = \pi_{i+}\pi_{+j}$

Expected frequency: $\mu_{ij} = n\pi_{ij} = n\pi_{i+}\pi_{+j}$

In practice, since $\{\pi_{i+}\}$and $\{\pi_{+j}\}$ are unknown, the estimated expected frequency using sample proportions is applied:

$$\hat{\mu}_{ij} = np_{i+}p_{+j} = n\frac{n_{i+}}{n}\frac{n_{+j}}{n} = \frac{n_{i+}n_{+j}}{n}$$

Test statistics:

$$X^2 = \sum\frac{(n_{ij} - \widehat{\mu_{ij}})^2}{\widehat{\mu_{ij}}}, \quad G^2 = 2\sum n_{ij}\log\left(\frac{n_{ij}}{\widehat{\mu_{ij}}}\right)$$

Approximately follows $\chi^2_{(I-1)(J-1)}$

Test of independence (Cont'd)

(Example) Party affliation by gender

성별	소속정당		
	민주당	공화당	무소속
여성	279	225	73
남성	165	191	47
합	444	416	120

$X^2 = 7.010 \sim \chi^2(df = 2)$: P-value=0.03

$G^2 = 7.003 \sim \chi^2(df = 2)$: P-value=0.03

Residual

The test statistic and p-value provide an overall assessment of evidence against the null hypothesis

If the null hypothesis is rejected, further detailed analysis is required.

Since the variance increases when $\hat{\mu}_{ij}$ is large, using only the raw residuals $(n_{ij} - \hat{\mu}_{ij})$is insufficient.

Adujsted residual:

Under Possion sampling,

$$Z_{ij} = \frac{n_{ij} - \widehat{\mu_{ij}}}{\sqrt{\widehat{\mu_{ij}}(1 - p_{i+})(1 - p_{+j})}}$$

For large sample sizes, under the null hypothesis, Z_{ij} approximately follows the standard normal distribution

Residual (Cont'd)

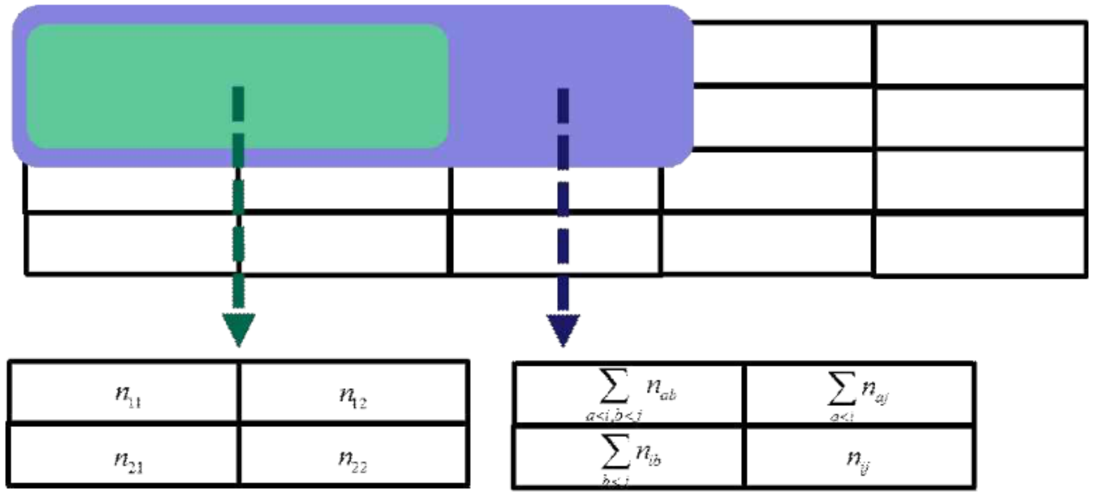
(Example) For the following survey results on party affiliation by gender, calculate the adjusted residuals

성별	소속정당		
	민주당	공화당	무소속
여성	279	225	73
남성	165	191	47
합	444	416	120

Partitioning of the Chi-square statistic

A chi-square statistic can be decomposed into several independent chi-square statistics with smaller degrees of freedom (this provides more detailed information about the association)

G^2 can be exactly partitioned, whereas X^2 cannot.



(Example) In a $2 \times J$ contingency table, the G^2 statistic can be partitioned into $(J-1)$ independent statistics, each with 1 degree of freedom

Partitioning of the Chi-square statistic (Cont'd)

(Example) Most Influential School of Psychiatric Thought and Ascribed Origin of Schizophrenia. Obtain the partitioned contingency table

School of Psychiatric Thought	Origin of Schizophrenia			합계
	Biogenetic	Environmental	Combination	
Eclectic	90	12	78	180
Medical	13	1	6	20
Psychoanalytic	19	13	50	82
합계	122	26	134	282

$$G^2 = 23.04 \sim \chi^2(df = 4)$$

Additional Notes on the Chi-square Test

Restrictions and Comparisons for X^2 and G^2

To use the chi-square distribution, the sample size must be large relative to the number of cells

X^2 converges more quickly than G^2 .When $n < 5IJ$, the approximation by G^2 is inadequate

When I or J is large, even if some expected cell frequencies are small, X^2generally performs reasonably well

Both statistics are invariant to the ordering of rows and columns, so they are regarded as methods suitable for analyzing nominal data.

Consideration

The chi-square test provides a measure of statistical significance of association, not a measure of the magnitude of association

Chi-square partitioning and residuals only provide information about which cells deviate from independence.

The odds ratio and relative risk can be used as measures of association strength in a 2 $\times$ 2 table, but they cannot be applied as measures of association strength in a general $I \times J$ table

Test for Independence for Ordinal Data

When the row variable X and the column variable Y are ordinal data

(Example) Father's and son's height

		아버지						합
		150~155	155~160	160~165	165~170	170~175	175~180	
아들	175~180					1	1	2
	170~175			1	2	2	2	7
	165~170		2	5	9	7	2	25
	160~165		2	10	16	7	1	36
	155~160	1	2	7	9	4	1	24
	150~155		1	2	2	1		6
합		1	7	25	38	22	7	100

Test for Independence for Ordinal Data (Cont'd)

Alternative Hypothesis of Linear Trend for Independence

Trend

A tendency for the level of Y to increase or decrease as the level of X increases

Linear trend (correlation): when the association of the trend is represented linearly

Row category scores: $u_1 \leq u_2 \leq \cdots \leq u_I$

Column category scores: $v_1 \leq v_2 \leq \cdots \leq v_J$

Test for Independence for Ordinal Data (Cont'd)

Pearson's product-moment correlation

$$r = \frac{\sum_{ij} u_i v_j n_{ij} - \left(\sum_i u_i n_{i+}\right)\left(\sum_j v_j n_{+j}\right)/n}{\sqrt{\sum_i u_i^2 n_{i+} - \left(\sum_i u_i n_{i+}\right)^2/n \quad \sum_j v_j^2 n_{+j} - \left(\sum_j v_j n_{+j}\right)^2/n}}$$

It is a measure of the strength of association for ordinal data

Range: $-1 \leq r \leq 1$

If the two variables are independent, then $r \approx 0$; the farther r is from 0, the more independence is in doubt

Under the null hypothesis (independence), approximately (Mantel, JASA 1963):

$$M = \sqrt{n-1}\,r \sim N(0,1) \;\rightarrow\; M^2 = (n-1)r^2 \sim \chi_1^2$$

Test for Independence for Ordinal Data (Cont'd)

(Example) Alcohol consumption of pregnant women and the presence of birth defects in newborns

알코올량	기형없음	기형있음	합계	기형비율(%)	수정잔차
0	17066	48	17114	0.28	-0.18
<1	14464	38	14502	0.26	-0.71
1-2	788	5	793	0.63	1.84
3-5	126	1	127	0.79	1.06
5<	37	1	38	2.63	2.71

Alcohol amount: represented as {0, 0.5, 1.5, 4, 7}

When alcohol amount is considered as a categorical variable: $X^2 = 12.1\ (p = 0.014)$,$G^2 = 6.5\ (p = 0.168)$

When alcohol amount is considered as an ordinal variable:

$M^2 = 6.570$: the null hypothesis of no correlation can be rejected

$M = -2.563$: as alcohol consumption increases, the proportion of newborns without defects

Test for Independence for Ordinal Data (Cont'd)

Detailed analysis

From the adjusted residuals: when alcohol consumption is low, the proportion of birth defects tends to be smaller than expected, but since the magnitude is small, it does not carry special significance

When alcohol consumption is 1 or higher, the values are relatively larger than expected
→ The greater the alcohol consumption, the higher the likelihood of birth defects

Selection of Category Scores

Relationship between category scores and M^2

In general, even if monotonic category scores are chosen in different ways, the results are similar

When the data are unbalanced, results may differ depending on the choice of category scores

In the above example, if $\{1,2,3,4,5\}$ are used as the category scores for alcohol consumption, then $M^2 = 1.83\ (p = 0.18)$

Mid-ranks

Assign ranks to all subjects and use the average ranks as category scores.

If adjacent categories have relatively small cell frequencies, they will have similar mid-ranks; in the birth defect example, the scores for 1–2, 3–5, and 5+ do not differ much.

In the above example, using mid-ranks yields $M^2 = 0.35\ (p = 0.55)$.

Spearman's ρ Test

Spearman's ρ Test

A nonparametric correlation coefficient

When both variables X and Y are ordinal, this is equivalent to M^2 using mid-ranks

Measures of Association for Ordinal Data

Gamma

A measure of association for ordinal data that does not require the use of category scores

Concordant pair: when two different subjects fall into cells (i, j) and (h, l) ,if $i < h$ and $j < l$, the two subjects are defined as a concordant pair

Discordant pair: when two different subjects fall into cells(i, j) and (h, l) ,if $i < h$and $j > l$, the two subjects are defined as a discordant pair

Definition: Let the number of concordant pairs be N_c and the number of discordant pairs be N_d .Then,

$$\gamma = \frac{\Pi_c - \Pi_d}{\Pi_c + \Pi_d}, \quad -1 \le \gamma \le 1$$

2	1
1	1

If the two variables are independent, then $\gamma = 0$

Measures of Association for Ordinal Data (Cont'd)

Kendall's τ_b

τ is a measure that accounts for ties in category scores

$T_X(T_Y)$: the number of subject pairs tied on the category scores of $X(Y)$

$$\tau_b = \frac{\Pi_c - \Pi_d}{\sqrt{(\Pi_c + \Pi_d + T_X)(\Pi_c + \Pi_d + T_Y)}}, \quad -1 \leq \tau_b \leq 1$$

If the two variables are independent, then $\tau_b = 0$

τ_b is useful when $I = J$; when $I \neq J$, Stuart's τ_c is more appropriate.

Measures of Association for Ordinal Data (Cont'd)

Stuart's τ_c

Let $m = \min(I, J)$ and n be the sample size

$$\tau_c = \frac{2m(\Pi_c - \Pi_d)}{n^2(m-1)}, \ -1 \le \tau_c \le 1$$

If the two variables are independent, then $\tau_c = 0$

τ_b is useful when $I = J$, while τ_c is more appropriate when $I \neq J$

Measures of Association for Ordinal Data (Cont'd)

Somer's D

When X is the explanatory variable and Y is the response variable

$$D = \frac{\Pi_c - \Pi_d}{\Pi_c + \Pi_d + T_Y}, \quad -1 \leq D \leq 1$$

If the two variables are independent, then $D = 0$

Tests such as the chi-square test, as well as measures like gamma, τ_b ,and τ_c ,do not distinguish between explanatory and response variables

Somers' D, however, is useful specifically when X is the explanatory variable and Y is the response variable

Measure of Association for Nominal Data

Proportion Reduction of Error (PRE)

If $V(Y)$ is a measure related to the variation of Y, then

$$\text{PRE} = \frac{V(Y) - E[V(Y|X)]}{V(Y)}, \quad 0 \le \text{PRE} \le 1$$

where

$$E[V(Y|X)] = \sum_{i=1}^{I} \pi_{i+} V(Y|X=i)$$

Interpretation:

Measure of Association for Nominal Data (Cont'd)

Types of PRE

(i) Uncertainty coefficient $V(Y) = \sum_{j=1}^{J} \pi_{+j} \log \pi_{+j}$

(ii) Lambda (λ) $V(Y) = 1 - \max\{\pi_{+1}, \dots, \pi_{+J}\}$

Properties of PRE

$PRE = 0$ means there is no association between X and Y

$PRE = 1$ means X and Y are not independent

Exact Tests for Small Samples

2×2 Contingency Table

a	b	$a+b$
c	d	$c+d$
$a+c$	$b+d$	n

Pearson's X^2 Statistic:

$$X^2 = \frac{n\,(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

When the sample size is small, the chi-square approximation is not accurate

Exact Tests for Small Samples (Cont'd)

Yates' Continuity Correction

$$X^2 = \frac{n\left(|ad - bc| - n/2\right)^2}{(a+b)(c+d)(a+c)(b+d)}$$

When the cell frequency is less than 3, even the continuity-corrected statistic does not perform well

Fisher's Exact Test

When $X \sim B(a+b, \pi) \perp Y \sim B(c+d, \pi)$

$$P(X=a|X+Y=a+c) = \frac{\binom{a+b}{a}\binom{c+d}{c}}{\binom{a+b+c+d}{a+c}}$$ (Hypergeometric distribution)

In a 2×2 contingency table, if the marginal totals are fixed:

The number of ways to choose a from $a+c$ is $\binom{a+c}{a}$

The number of ways to choose b from $b+d$ is $\binom{b+d}{b}$

Total number of possible cases: $\binom{a+c}{a}\binom{b+d}{b}$

When the marginal totals are fixed, the probability that cell (1,1) contains a is:

$$P(n_{11}=a|n_{1+}=a+c, n_{+1}=a+b) = \frac{\binom{a+c}{a}\binom{b+d}{b}}{\binom{n}{a+b}}$$

$$= \frac{(a+b)!(a+c)!(c+d)!(b+d)!}{a!b!c!d!n!}$$

Fisher’s Exact Test (Cont’d)

(Example) The responses of Psychotics and Neurotics to suicide are given below. Perform an exact test

자살충동	psychotic	neurotics	합
예	2	6	8
아니오	18	14	32
합	20	20	40

Noncentral Hypergeometric Distribution

x	$n-x$	n
y	$m-y$	m
$x+y$	$n+m-(x+y)$	$n+m$

If the odds ratio is θ, the probability mass function is given by:

$$P(X=x,\, Y=y \mid X+Y=t) = \frac{\binom{n}{x}\binom{m}{t-x}}{\sum_x \binom{n}{x}\binom{m}{t-x}\theta^x}\theta^x$$

When the odds ratio is 1, this reduces to the hypergeometric distribution

The mean and variance of the hypergeometric distribution are as follow

$$E(X) = \frac{nt}{n+m},\, var(X) = \frac{nmt(n+m-t)}{(n+m)^2(n+m-1)}$$

Summary

(Example) What is the appropriate test statistic in each of the following cases?

<경우 1>

	Y_1	Y_2	Y_3	Y_4	계
$X=1$	10	20	40	50	120
$X=2$	50	30	10	20	110
계	60	50	60	70	230

<경우 2>

	Y_1	Y_2	Y_3	Y_4	계
$X=1$	1	2	4	5	12
$X=2$	5	3	1	2	11
계	6	5	6	7	23

<경우 3>

	Y_1	Y_2	Y_3	Y_4	계
$X=1$	1	2	40	50	93
$X=2$	5	3	10	20	38
계	6	5	60	70	131

Partial Association

Notation: Y= response variable, X= explanatory variable, Z= control variable

Control variable: When studying the effect of the explanatory variable on the response variable, there may be other variables that influence this relationship

If the control variables are not held constant at each level of the explanatory and response variables, problems may arise because the relationship between X and Y can be affected

Three-way Table

A table in which data are cross-classified by the levels of three variables, and the observed frequencies are recorded

(Example) A study of the relationship between the defendant's race, the victim's race, and the death penalty verdict

Radelet (1981): Results of trials for 326 defendants indicted for murder in 20 Florida counties between 1976 and 1977

피해자	피의자	사형판결	
		예	아니오
백인	백인	19	132
	흑인	11	52
흑인	백인	0	9
	흑인	6	97

Partial Table

A two-dimensional cross-classification of X and Y at each level of Z

Since it shows the relationship between $X$and Y at a fixed level of Z, the effect of Z is removed

(Example) If the victim is treated as the control variable

피해자	피의자	사형판결	
		예	아니오
백인	백인	19	132
	흑인	11	52
흑인	백인	0	9
	흑인	6	97

Marginal Table

A two-way contingency table obtained by summing the cell frequencies in the same position across the partial tables

It represents the relationship between $X$and Y while ignoring the effect of Z

(Example) Marginal table of defendants and death penalty verdicts:

피의자	사형판결		합	사형비율
	예	아니오		
백인	19	141	160	11.9
흑인	17	149	166	10.2

According to the marginal table, the proportion of death penalty verdicts is about 11.9% for White defendants and about 10.2% for Black defendants, indicating that the rate is slightly lower for Black defendants

Conditional Association

The association observed in the partial table when the level of Z is fixed

Since the conditional association in the partial tables can differ from the association in the marginal table, analyzing only the marginal table in multi-dimensional contingency tables can lead to incorrect conclusions: Simpson's paradox.

Conditional Association

(Example) Data on Defendant's Race, Victim's Race, and Death Penalty Verdicts

Trial outcomes of 326 defendants indicted for murder in Florida counties

피해자	피의자	사형판결		사형 비율
		예	아니오	
백인	백인	19	132	12.6
	흑인	11	52	17.5
흑인	백인	0	9	0.0
	흑인	6	97	5.8

When the victim was White, the proportion of death penalty verdicts was higher for Black defendants

When the victim was Black, the proportion of death penalty verdicts was also higher for Black defendants

When the victim variable was stratified and analyzed, the conclusion was the exact opposite of that drawn from the marginal table

Conditional Association (Cont'd)

Table. Odds Ratios among Death Penalty (P), Victim (V), and Defendant (D)

연관성		P-D	P-V	D-V
주변표		1.18	2.71	25.99
부분표	Level 1	0.67	2.80	22.04
	Level 2	0.79	3.29	25.90

The odds of a White victim being killed by a White defendant are much higher than the odds of a White victim being killed by a Black defendant (25.99, 22.04, 25.90)

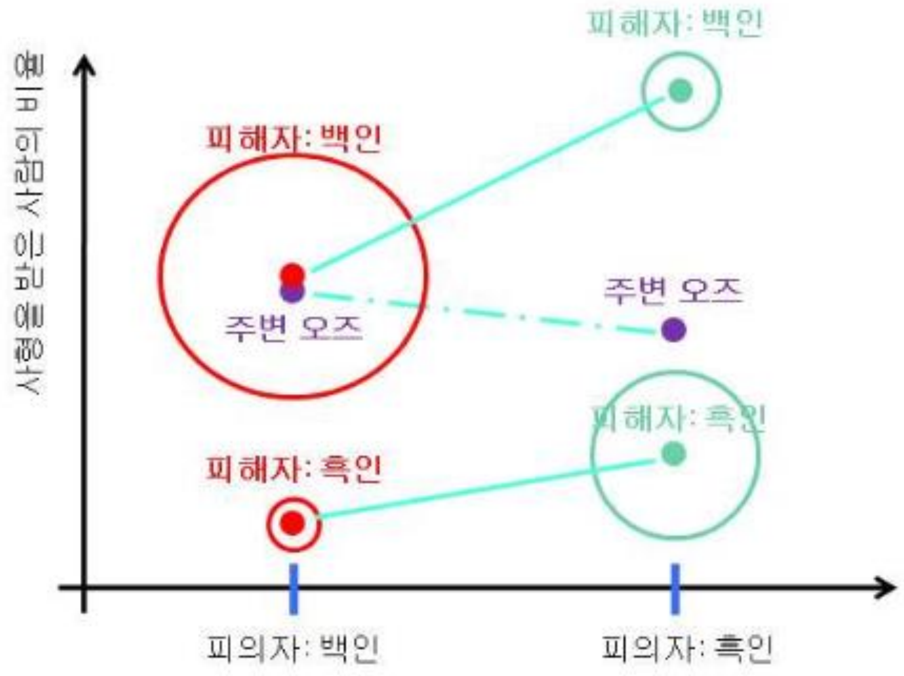
There is a very strong association between the victim's race and the defendant's race

When the victim is White, the odds of a death penalty verdict are higher compared to the odds for Black victims (2.71, 2.80, 3.29)

The victim's race is a major factor in the death penalty verdict

Conditional Association (Cont’d)

Conclusion: Whites tend to kill other Whites more frequently, and since death penalty verdicts are higher when the victim is White, the marginal table shows that death penalty rates appear higher for White victims.



Conditional Odds Ratio and Marginal Odds Ratio

Conditional Odds Ratio

$\{n_{ijk}\}$:observed frequency in cell (i, j, k)

$\{\mu_{ijk}\}$: expected frequency in cell (i, j, k)

In a $2 \times 2 \times K$ table, the conditional association between $X$and Yat a fixed level k of Z

$$\theta_{XY(k)} = \frac{\mu_{11(k)}\mu_{22(k)}}{\mu_{12(k)}\mu_{21(k)}} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}}$$

The odds ratio in the k-th partial table (conditional odds ratio)

Sample conditional odds ratio:

$$\hat{\theta}_{XY(k)} = \frac{n_{11k}n_{22k}}{n_{12k}n_{21k}}$$

Conditional Odds Ratio and Marginal Odds Ratio (Cont'd)

Marginal Odds Ratio

Calculated by ignoring the control variable: $\theta_{XY} = \frac{\mu_{11+}\mu_{22+}}{\mu_{12+}\mu_{21+}}$

Sample marginal odds ratio: $\hat{\theta}_{XY} = \frac{n_{11+}n_{22+}}{n_{12+}n_{21+}}$

(Example) Odds ratio between defendant and death penalty verdict:

$$\hat{\theta}_{XY} = \frac{19 \times 149}{141 \times 17} = 1.18$$

$$\hat{\theta}_{XY(\text{백인})} = \frac{19 \times 52}{11 \times 132} = 0.68$$

$$\hat{\theta}_{XY(\text{흑인})} = \frac{0.5 \times 9.5}{6.5 \times 97.5} = 0.79$$

Conditional Independence and Marginal Independence

Conditional Independence

If X and Y are independent in each partial table, then given Z, X and $Y$are conditionally independent:

$$P(X=x, Y=y|Z=z) = P(X=x|Z=z)P(Y=y|Z=z)$$

$$\pi_{ijk} = \frac{\pi_{i+k}\pi_{+jk}}{\pi_{++k}}$$

All conditional odds ratios equal 1:

$$\theta_{XY(k)} = 1, \ \forall k$$

Conditional Independence and Marginal Independence (Cont'd)

Marginal Independence

X and Y are independent when ignoring the control variable Z:

$$\theta_{XY} = 1$$

Conditional independence of X and Y at all levels of Z does not necessarily imply marginal independence.

Conditional Independence and Marginal Independence (Cont'd)

(Example) Relationship between Gender and Income by Major

전공	성별	소득	
		하	상
교양학문	여자	0.18	0.12
	남자	0.12	0.08
과학, 공학	여자	0.02	0.08
	남자	0.08	0.32
합	여자	0.20	0.20
	남자	0.20	0.40

Odds ratios by major classification:

$$\theta_{\text{교양}} = \frac{0.18 \times 0.08}{0.12 \times 0.12} = 1.0, \ \theta_{\text{과학공학}} = \frac{0.02 \times 0.32}{0.08 \times 0.08} = 1.0$$

When the major is given, income level and gender satisfy conditional independence

Conditional Independence and Marginal Independence (Cont'd)

When the major is not considered, the two variables are not independent

$$\theta = \frac{0.20 \times 0.40}{0.20 \times 0.20} = 2$$

In particular, the ratio of high-income to low-income among men is twice that of women, showing that men are more likely to have higher income levels

$X - Z$ odds ratio: the marginal odds in Science/Engineering are 6 times higher for men than for women

$Y - Z$ odds ratio: the marginal odds of high income are 6 times higher in Science/Engineering compared to Liberal Arts

Science and Engineering majors are relatively male-dominated, and these majors are shown to have relatively higher income

Homogeneous Association

If the odds ratio values are the same across all levels of the control variable Z, then X–Y are said to have homogeneous association:

$$\theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(K)}$$

This means that the effect of X on Y is the same across all levels of Z, and thus the X–Y homogeneous association can be summarized by a single number

Note: X–Y homogeneous association does not imply conditional independence of X and Y

Conditional independence requires the conditional odds ratio of X–Y to be fixed at 1

(Example) X =drinking (yes, no), Y =liver cancer (yes, no), Z =age (30s, 40s, 50s)

$$\theta_{XY(30)} = 1.15,\ \theta_{XY(40)} = 2.30,\ \theta_{XY(50)} = 5.72$$

In the 30s group, drinking does not have much effect on liver cancer, but as age increases, the probability of developing liver cancer among drinkers shows a clear increasing trend

Homogeneous Association in an $I \times J \times K$ Table

When, at each level of Z, the conditional odds ratio determined by any two levels of X and Y is the same.

Relationship between Homogeneous Association and Regression Models

(i) $y_{ij} = \beta_0 + \beta_1 x_{i1} + \epsilon_{ij}$, β_1 explains the marginal association between X_1 and Y

(ii) $y_{ij} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_{ij}$

If there is homogeneous association between X_1 and Y, then β_1 explains the conditional association of X_1–Y

If there is homogeneous association between X_2 and Y, then β_2 explains the conditional association of X_2–Y

Homogeneous Association in an $I \times J \times K$ Table

$$\text{(iii)} \quad y_{ij} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_1 \beta_2 x_{i1} x_{i2} + \epsilon_{ij}$$

In this case, homogeneous association between X_1 and Y is not satisfied

Note: Homogeneous association is a symmetric property applied to a pair of variables when controlling for another variable. When this property holds, it implies that there is no interaction between the variables.